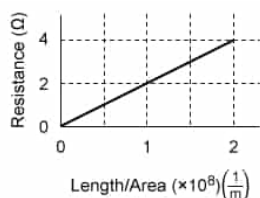


The dependence of a metal's resistance on its length per unit area is shown.



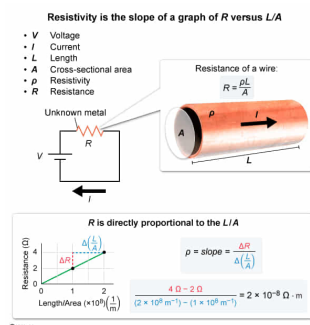
What is the resistivity of the metal?

- ☐ A. $0.5 \times 10^{-8} \Omega \cdot \text{m}$ [6%]
☒ B. $2 \times 10^{-8} \Omega \cdot \text{m}$ [83%]
☐ C. $4 \times 10^{-8} \Omega \cdot \text{m}$ [6%]
☐ D. $8 \times 10^{-8} \Omega \cdot \text{m}$ [2%]

Correct

83% answered correctly

Explanation:



Resistivity ρ is a property that characterizes a material's intrinsic ability to impede the flow of charge. The **resistance** R in a sample of a material is **directly proportional** to the product of the material's resistivity and the sample's length L and is **inversely proportional** to the sample's cross-sectional area A :

$$R = \frac{\rho L}{A}$$

In this question, the graph shows measured resistance of a metal with respect to the length to area ratio. The **best-fit line** indicates a directly proportional relationship between the quantities measured on each axis and can be related to the resistivity equation as:

$$y = m \cdot x$$

$$R = \rho \cdot \left(\frac{L}{A} \right)$$

Therefore, R is directly proportional to the quantity L/A , and the **slope** is represented by the ρ . To determine ρ , the slope can be calculated between two points that lie on the best-fit line. The resistance has a value of 2Ω when the length to area ratio is $1 \times 10^8 \text{ m}^{-1}$ and a value of 4Ω when the length to area ratio is $2 \times 10^8 \text{ m}^{-1}$. Hence, the resistivity is given by:

$$\rho = \frac{\text{rise}}{\text{run}} = \frac{\Delta R}{\Delta \left(\frac{L}{A} \right)}$$

$$\rho = \frac{4 \Omega - 2 \Omega}{2 \times 10^8 \text{ m}^{-1} - 1 \times 10^8 \text{ m}^{-1}}$$

$$\rho = \frac{2 \Omega}{1 \times 10^8 \text{ m}^{-1}} = 2 \times 10^{-8} \Omega \cdot \text{m}$$

Therefore, the resistivity of the metal is $2 \times 10^{-8} \Omega \cdot \text{m}$.

(Choices A, C, and D) These answer choices are multiples of $\frac{1}{4}$, 2, and 4 off, respectively, from the correct answer, most likely due to incorrectly dividing the change in the length to area ratio by the change in the resistance (or by misplacing numbers within the slope formula).

Educational objective:

The resistivity of a conductor is an intrinsic property that measures the conductor's ability to impede charge flow. Resistivity equals the slope of a graph of resistance versus length per unit area.

Time spent: 0 sec
 Subject:
 Physics

QID: 403997
 System:
 Electrostatics and Circuits